PranaV- H
(192525420)

p SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO5:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3 **Total Marks:30**

13

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

	Criteria	Weightage (%)		Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
A	Understanding of Problem & Constraints	20%	2	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
B	Problem Decomposition	15%	1.5	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
C	Application of Constraints in Solution	20%	2	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
D	Logical Reasoning & Strategy	15%	1.5	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
E	Solution Accuracy & Feasibility	15%	1.5	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
F	Creativity & Innovation	5%	.5	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
G	Clarity of Presentation	10%	-1	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

36.5

	A	B	C	D	E	F	G	
Q1	1.5	1	1.5	.5	1.5	.5	1	7.5
Q2	1	1	1.5	1.5	1.5	.5	1	8
Q3	1.5	1	2	1	.5	.5	.5	7
Q4	1.5	1	2	1	1	.5	.5	7.5
Q5	1.5	.5	1.5	1	1	.5	.5	6.5

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Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

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ANSWERS

Constraint-Based Problem Solving – Solutions to Questions 1, 2 & 3

Question 1: A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.

Answer 1

1. Understanding of the Problem & Constraints

The core problem is to provide reliable, secure, and cost-effective connectivity to a remote rural branch office that has no access to wired broadband (fibre/DSL/leased line). The solution must simultaneously satisfy four constraints:

- Bandwidth: must handle cloud application traffic and real-time video conferencing without lag.
- Reliability: business operations must not be interrupted even if the primary link fails.
- Cost: the company has a limited budget, so capital and recurring costs must be minimised.
- Scalability: the design must allow the office to grow (more staff, more bandwidth-hungry applications) without a complete redesign.

2. Problem Decomposition

The problem can be broken into four logical sub-problems, each solved independently and then integrated:

1. Choice of last-mile internet connectivity for a location with no wired infrastructure.
2. Selection of networking/CPE (Customer Premises Equipment) devices to terminate and manage that link.
3. Design of a security mechanism to protect data travelling to headquarters (HQ).
4. A redundancy/failover plan and a scalability roadmap.

3. Application of Constraints in the Solution

Primary Internet Connection: 4G LTE / 5G Fixed Wireless Access (FWA), or VSAT (satellite) where cellular coverage is weak.

- 4G/5G FWA is chosen as the primary link because it needs no cabling, has moderate cost, and provides 10–50 Mbps – sufficient for cloud apps and HD video conferencing (addresses bandwidth + cost).
- VSAT is recommended only as a fallback in areas with zero cellular coverage, since it is more expensive and has higher latency.

Secondary/Backup Connection: A second, independent 4G/LTE SIM from a different carrier (or a low-cost VSAT backup) configured for automatic failover – this directly satisfies the "uninterrupted business operations" requirement (reliability constraint) without the high cost of a second wired leased line.

Networking Devices:

- SD-WAN capable router / dual-SIM LTE router – aggregates or fails over between the two WAN links automatically and prioritises (QoS) video-conferencing and VoIP traffic over bulk downloads.
- Next-generation firewall (NGFW) with built-in VPN – combines routing, firewall, and VPN termination in one low-cost appliance, ideal for a small branch (cost constraint).
- A managed Layer-2 switch and a Wi-Fi 6 access point complete the LAN for office staff.

Security Mechanisms:

- Site-to-Site IPsec VPN tunnel between the branch router and HQ firewall encrypts all inter-office traffic.
- Zero Trust / MFA for access to cloud applications, since a mobile/wireless WAN link is inherently more exposed than a private wired circuit.
- Cloud-based DNS filtering and endpoint antivirus to protect against malware, since the branch has no on-site IT staff.

4. Logical Reasoning & Strategy

The reasoning follows a prioritisation strategy: because wired broadband is physically unavailable, wireless WAN is the only feasible last-mile option; among wireless options, cellular (4G/5G) is preferred over VSAT for cost and latency, and VSAT is retained only as a redundancy option in poor-coverage zones. Reliability is engineered through dual-WAN failover rather than a single expensive high-SLA circuit, which keeps cost low while still meeting the "uninterrupted operations" requirement. Security is layered (VPN + MFA + filtering) so that no single control failure compromises HQ systems.

5. Solution Accuracy & Feasibility

This design is technically proven – SD-WAN over dual-LTE with IPsec VPN is a widely deployed pattern for exactly this scenario (retail branches, oil-field sites, rural clinics). It is feasible within a limited budget because it avoids expensive leased-line installation (which can take months and cost lakhs of rupees in rural terrain) and instead uses commodity LTE/5G hardware with monthly data plans.

6. Creativity & Innovation

An innovative addition is the use of SD-WAN policy-based routing: video conferencing and VoIP packets are automatically steered onto whichever of the two WAN links currently has the lowest latency/jitter, while bulk file-sync traffic is throttled — squeezing maximum usable performance out of low-cost wireless links rather than simply buying more bandwidth.

7. Scalability for the Future

- Add more SIM/carrier links or upgrade to 5G as coverage improves, without replacing the SD-WAN router.
- Increase cloud application capacity purely through subscription upgrades (no on-site hardware change).
- Migrate to fibre/leased line later, if it becomes available, by simply adding it as another SD-WAN uplink.

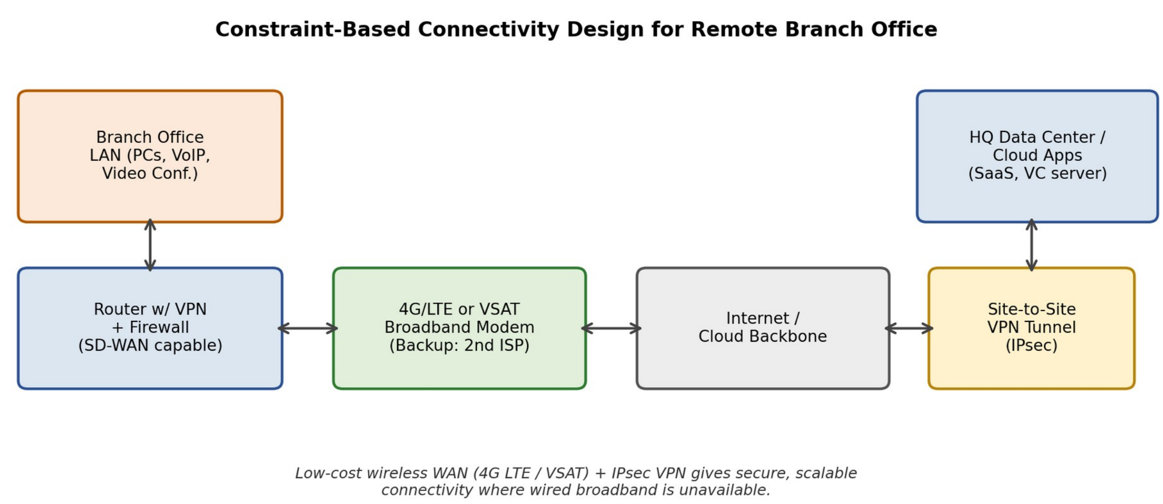

Figure 1: Constraint-based connectivity design for the remote branch office.

Question 2: A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

Answer 2

1. Understanding of the Problem & Constraints

The university needs Wi-Fi for 3,000+ concurrent users across four very different zones (classrooms, labs, library, hostels), each with different user density and usage patterns. The key constraints are:

- Capacity: many simultaneous users/devices per access point (AP).
- Interference: overlapping Wi-Fi channels between adjacent APs must be minimised.
- Security: only authorised students/staff should get network access.
- Cost: a limited number of APs can be purchased and deployed.

2. Problem Decomposition

1. Estimate user density per zone and calculate the number of APs needed per building.
2. Decide AP placement/topology to maximise coverage per device (cost efficiency).
3. Select frequency bands and channel plan to reduce co-channel interference.
4. Choose an authentication and access-control scheme.

3. Application of Constraints in the Solution

Access Point Selection & Placement:

- Use enterprise-grade dual-band 802.11ax (Wi-Fi 6) APs – Wi-Fi 6 supports far more simultaneous client associations per AP (OFDMA/MU-MIMO) than older standards, reducing the total number of APs needed (directly addresses the limited-AP-count constraint).
- Place APs using a cellular/honeycomb layout with 60–80% cell overlap only at the edges, mounted on ceilings at regular intervals – typically one AP per 500–600 sq. ft. in high-density areas (classrooms/labs/library reading rooms) and one AP per 2–3 hostel rooms cluster.
- In large lecture halls and libraries, prefer 2–3 APs with lower transmit power over 1 high-power AP, since high-density, low-power cells give better per-user throughput than one powerful AP.

Frequency Band Selection:

- 5 GHz (and 6 GHz where hardware supports Wi-Fi 6E) is used as the primary band for data-heavy areas (labs, library) because it has more non-overlapping channels (36, 40, 44...161) and less interference.
- 2.4 GHz is retained only for legacy IoT devices and as a fallback for range in hostels, using only channels 1, 6, 11 to avoid overlap.
- Channel width is kept at 20–40 MHz in dense areas (not 80/160 MHz) to fit more non-overlapping channels into the limited spectrum, reducing co-channel interference among the limited AP count.

Authentication Methods:

- WPA2/WPA3-Enterprise with 802.1X and a central RADIUS server tied to the university's student/staff directory (LDAP/Active Directory) – each user gets unique credentials, avoiding a single shared password (security constraint).
- A separate, rate-limited guest SSID (captive portal) isolates visitor traffic from the academic network.
- A Wireless LAN Controller (WLC) centrally manages all APs – enabling automatic channel/power adjustment (RRM) and reducing management overhead, which is important since a limited IT team is supporting a large network.

4. Logical Reasoning & Strategy

The strategy is to maximise "users served per AP" rather than simply buying more APs, since the budget caps AP count. This is achieved by (a) choosing a Wi-Fi standard with higher per-AP client capacity, (b) using a high-density/low-power cellular layout instead of few high-power APs, and (c) offloading legacy/IoT devices to 2.4 GHz so 5 GHz capacity is reserved for high-demand student devices. Security is centralised via RADIUS/802.1X so it scales to thousands of users without per-device configuration.

5. Solution Accuracy & Feasibility

This design mirrors real-world enterprise campus Wi-Fi deployments and is fully feasible: a mid-sized university campus (four buildings) can be adequately covered by roughly 60–80 Wi-Fi 6 APs under a WLC, which comfortably supports 3,000+ concurrent users when combined with proper channel planning and RRM.

6. Creativity & Innovation

An innovative element is dynamic band-steering combined with airtime fairness policies on the WLC: devices capable of 5 GHz are automatically nudged onto that band, and airtime is fairly divided so that one bandwidth-heavy user (e.g., large file download in the library) cannot starve others in the same cell — improving perceived performance without adding a single extra AP.

7. How the Solution Balances Coverage, Performance, Security & Cost

- Coverage: cellular AP layout with overlap only at cell edges ensures no dead zones across all four zones.
- Performance: Wi-Fi 6 + dual-band + RRM channel planning keeps throughput high despite user density.
- Security: centralised 802.1X/RADIUS + isolated guest network protects academic data with minimal per-user administrative effort.
- Cost: higher-capacity APs per unit, deployed in an optimised layout, meet demand within the limited AP-count budget instead of overspending on hardware.

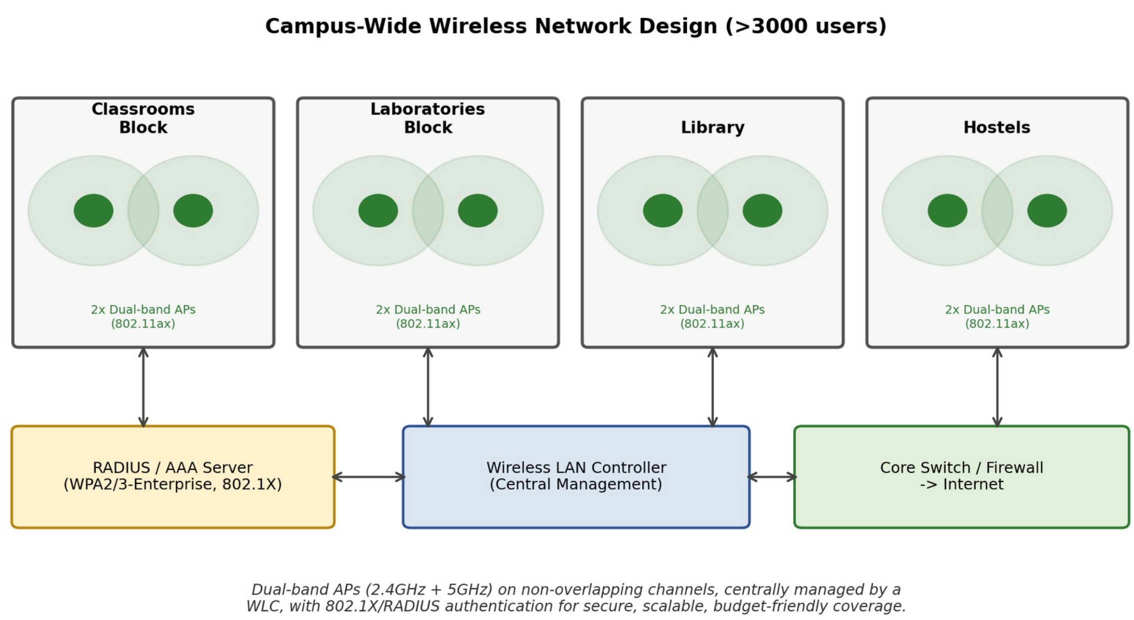

Figure 2: Campus-wide wireless network design showing AP placement, frequency plan, and authentication architecture.

Question 3: A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

Answer 3

1. Understanding of the Problem & Constraints

The financial institution must protect highly sensitive customer and banking data while still letting employees work efficiently and customers use online banking. The design must jointly satisfy:

- Security: strong protection against external attackers and insider threats.
- Performance: security controls must not introduce unacceptable latency.
- Compliance: must meet regulatory/security standards (e.g., PCI-DSS, RBI/ISO 27001-type guidelines) that mandate segmentation, logging, and access control.
- Budget: cannot significantly raise operational costs.

2. Problem Decomposition

1. Segment the network so that a breach in one zone cannot spread to another (containment).
2. Deploy firewalls at the correct boundary points (perimeter and internal).
3. Choose authentication mechanisms appropriate to each user/service type.
4. Add intrusion detection/prevention and monitoring for compliance and early threat detection.

3. Application of Constraints in the Solution

Network Segmentation:

- A DMZ (De-Militarized Zone) hosts all public-facing services (online banking web servers, reverse proxy) so that customers and the internet never touch the internal network directly.
- Internal VLAN segmentation separates the network into at least four zones: customer/database servers (VLAN 10), employee workstations (VLAN 20), online-banking application servers (VLAN 30), and an admin/monitoring zone (VLAN 40) — this limits lateral movement if any one segment is compromised (security constraint) while VLANs are a low-cost, switch-based technique (budget constraint).

Firewall Deployment:

- An external (perimeter) firewall with IDS/IPS sits between the internet and the DMZ, filtering all inbound/outbound traffic.
- An internal firewall acts as a segmentation gateway between the DMZ/core switch and the sensitive VLANs, enforcing that, for example, employee workstations cannot directly query the customer database — only the application-tier servers can, and only on required ports (least-privilege access, supports compliance).

Authentication Methods:

- Multi-Factor Authentication (MFA) for all employee access to customer data and admin systems, and for customers' online banking logins – directly satisfies strict security-policy/compliance requirements.
- Role-Based Access Control (RBAC) tied to Active Directory/LDAP ensures employees only see data relevant to their role (tellers vs. loan officers vs. IT admins).
- Certificate-based authentication (802.1X) for devices connecting to internal VLANs, preventing unauthorised device connections.

Intrusion Detection/Prevention:

- A Network IDS/IPS is placed both at the perimeter and at the internal segmentation gateway to detect known attack signatures and anomalous traffic in real time.
- Centralised logging via a SIEM (Security Information and Event Management) tool aggregates logs from firewalls, IDS, and servers for compliance audit trails and faster incident response.

4. Logical Reasoning & Strategy

The reasoning follows defence-in-depth: instead of relying on one strong perimeter firewall (single point of failure), security is layered — perimeter firewall, DMZ isolation, internal firewall/VLAN segmentation, MFA/RBAC, and IDS/IPS with SIEM monitoring. This ensures that even if an outer layer is breached, the sensitive VLANs (especially customer data) remain protected, satisfying both the security and compliance constraints. Performance is preserved by using VLANs and switch-based segmentation (which operate at line-rate) rather than routing all internal traffic through resource-heavy inspection, and by placing deep inspection (IDS/IPS) only at the key boundary points rather than on every internal link — keeping cost and latency low.

5. Solution Accuracy & Feasibility

This layered DMZ + VLAN-segmentation + MFA + IDS/IPS + SIEM architecture is the accepted industry-standard design for banks and financial institutions and directly maps to common compliance frameworks (PCI-DSS segmentation and monitoring requirements). It is feasible because it largely reuses existing switches (VLAN capability) and adds firewalls/IDS only at a small number of boundary points, making it both technically sound and budget-realistic.

6. Creativity & Innovation

An innovative refinement is applying micro-segmentation with next-generation firewall (NGFW) application-aware rules at the internal gateway — rather than allowing/blocking by IP and port alone, the firewall inspects the specific banking application protocol, so even a compromised workstation cannot issue arbitrary database commands, adding a strong layer of protection at no extra hardware cost beyond the existing internal firewall.

7. How the Design Balances Security, Performance, Compliance & Budget

- Security: DMZ + VLAN segmentation + MFA + IDS/IPS gives multiple independent barriers against breaches.
- Performance: switch-based VLAN segmentation is line-rate; deep inspection is applied only at key chokepoints, not on every packet everywhere.
- Compliance: segmentation, MFA, and centralised SIEM logging directly satisfy audit and regulatory requirements for financial data protection.
- Budget: the design reuses existing switching infrastructure for segmentation and adds firewalls/IDS at only two boundary points, avoiding costly per-device security appliances.

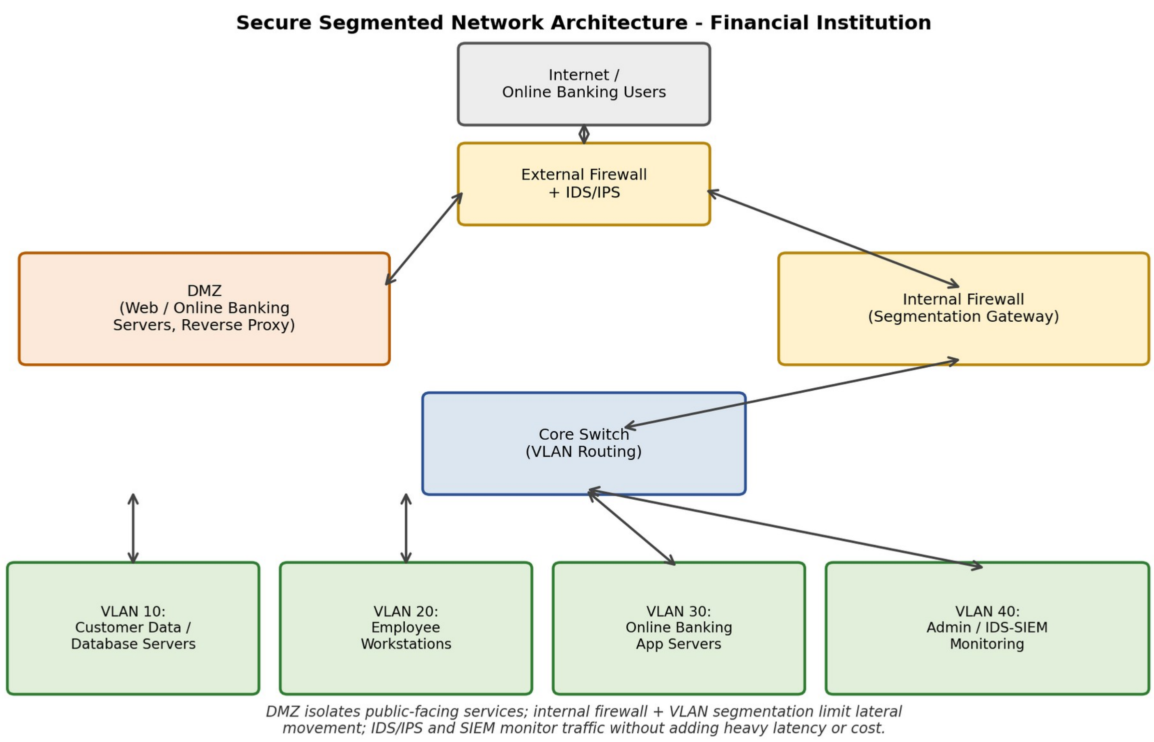

Figure 3: Secure segmented network architecture for the financial institution.